

PRINTF

prints formatted output

PRINTF

A little bit about Printf

```
func main() {  
    var brand string  
    fmt.Printf("%q\n", brand)  
}
```



PRINTF

A little bit about Printf

```
func main() {  
    var brand string  
    fmt.Printf("%q\n", brand)  
}
```

```
$ go run main.go
```

```
""
```


PRINTF

A little bit about Printf

```
func main() {  
    var brand = "Google"  
  
    fmt.Printf("%q\n", brand)  
}
```

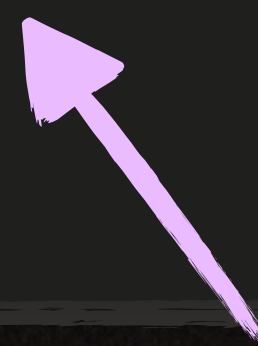
```
$ go run main.go
```

```
"Google"
```


PRINTF

A little bit about Printf

```
func main() {  
    var brand = "Google"  
  
    fmt.Printf("%s\n", brand)  
}
```



```
$ go run main.go
```

Google

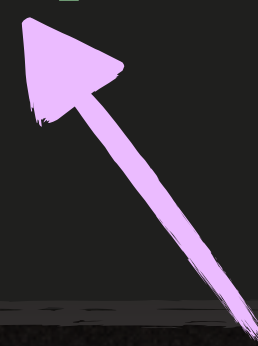


"Google"

PRINTF

A little bit about Printf

```
func main() {  
    var brand = "Google"  
  
    fmt.Printf("%q\n", brand)  
}
```



```
$ go run main.go
```

```
"Google"
```


formatting text

determines "what" and "how" to print



```
fmt.Printf("%q\n", brand)
```



replacer value(s)

*replaces the verbs inside
the formatting text*

verb

getting replaced by the passed value



```
fmt.Printf("%q\n", brand)
```



value(s)

*replaces the verbs inside
the formatting text*


```
fmt.Printf("%q\n", brand)
```



escape sequence
prints a newline

PRINTLN vs PRINTF

total: 2350 **success:** 543 / 433

PRINTLN

Please assume that "ops", "ok" and "fail" variables are declared already

```
fmt.Println(  
    "total:", ops, "success:", ok, "/", fail,  
)
```

total: 2350 **success:** 543 / 433

PRINTF

```
fmt.Printf(  
    "total: %d success: %d / %d\n",  
    ops, ok, fail,  
)
```

total: 2350 success: 543 / 433

ESCAPE SEQUENCES

Allow you to represent special characters

ESCAPE SEQUENCES

They give special meanings to characters

```
func main() {  
    fmt.Println( "hihi" )  
}
```

```
$ go run main.go  
hihi
```


ESCAPE SEQUENCES

They give special meanings to characters

```
func main() {  
    fmt.Println("hi\nhi")  
}
```

Go **interprets**
string values

```
$ go run main.go
```

```
hi
```

```
hi
```


\n is for printing a new line

\n is an escape sequence

fmt.Println("hi\nhi")

\n is one character
this string's length is 5 characters

\n



When a character **starts** with a

it might be an **escape sequence**

\\



\

\n



new
line

\"



"

"hi\\n\"hi\""

ESCAPE SEQUENCES

They give special meanings to characters

```
func main() {  
    fmt.Println("hi\\n\\hi\\")  
}
```

Go **interprets**
string values

```
$ go run main.go
```

```
hi\n"hi"
```


PRINTING TYPES

You can use `printf` to print the *types of values* (variables, constants, so on...)

PRINTING TYPES

You can use *printf* to print the *types of values* (variables, constants, so on...)

```
package main
import "fmt"

func main() {
    var speed int
    var heat float64
    var off bool
    var brand string

    fmt.Printf("%T\n", speed)
    fmt.Printf("%T\n", heat)
    fmt.Printf("%T\n", off)
    fmt.Printf("%T\n", brand)
}
```

```
$ go run main.go
```

int

float64

bool

string



s u m m a r y

format specifier



```
fmt.Printf(  
    "Your age is %d, my age is %d.",  
    age, yourAge,  
)
```


verbs

%s %q %d %f %.2f %t %v %T

`%v`
(swiss-knife)

it doesn't check for the type-safety
but it can print any value

printing types with %T

escape sequences

`\n`

`\"`

`\\`